

MSC FINANCE & BIG DATA

Optimization Methods with Applications to Finance • 2627_FI_5_3_073_E

Linear and Combinatorial Optimization

From the corner point method to NP-hardness

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Sessions 2 and 3 • Workshops • 2×180 min

What this chapter does

Part I — Linear programming

1. What makes a problem linear
2. Formulating: tables and chairs
3. Graphical solution, corner points
4. Four special cases
5. Sensitivity analysis
6. Duality
7. **Excel Solver**
8. Portfolio selection as an LP

Part II — Combinatorial optimization

9. When variables must be integers
10. The knapsack problem
11. **P, NP and NP-hardness**
12. Greedy, and why it fails
13. Relaxation and bounds
14. Branch and bound
15. Modelling with 0–1 variables
16. TSP and the factorial wall

Takeaway

Chapter 1 ended on a promise: an affine objective over a polytope attains its optimum at a *vertex*. Part I cashes that promise in. Part II asks what happens when we forbid the fractions — and finds that the price is **exponential**.

The fil rouge

Fil rouge — NEOMAMA Brokerage

NEOMAMA Brokerage has been instructed by a client to invest \$250,000 obtained from the sale of land holdings. The client trusts the house, but imposes her own guidelines on how the funds are distributed.

The same client, and the same \$250,000, come back in every chapter:

Chapter	What we add	Model class
2 — Part I	Diversification rules, return maximization	LP
2 — Part II	“At most K holdings”, all-or-nothing tickets	0–1 IP
3	Risk as <i>variance</i> , not a linear index	QP, KKT, Newton
4	Return <i>and</i> risk <i>and</i> ESG, simultaneously	Multi-objective
5	Her attitude to risk, made explicit	Utility, dominance

Everything you build this afternoon survives into the next four sessions.

Part I — Linear programming

What is a linear program?

Definition 1 (LP in standard form)

$$\begin{array}{ll}\text{maximize} & t^t c x \\ \text{subject to} & A x \leq b \\ & x \geq 0\end{array}$$

Four properties every LP shares:

- an objective to **maximize or minimize**;
- **constraints** that limit how far we pursue it;
- **alternative courses of action** to choose between;
- objective and constraints expressed as **linear** equations or inequalities.

Five modelling assumptions

1. **Certainty** — coefficients are known and fixed.
2. **Proportionality** — doubling an activity doubles its contribution and its resource use.
3. **Additivity** — the total is the sum of the parts; no interaction terms.
4. **Divisibility** — fractional solutions are allowed.
5. **Non-negativity** — $x \geq 0$.

Where it breaks

Assumption 4 is the one Part II removes. Assumption 1 is the one Chapter 5 removes. Each removal has a price.

Formulating: the carpentry problem

Tables and chairs

A workshop makes tables and chairs. Each table needs 4 hours of carpentry and 2 hours of painting; each chair needs 3 hours of carpentry and 1 hour of painting. There are 240 carpentry hours and 100 painting hours available. A table yields \$7 profit, a chair \$5. What should be produced?

Step 1 — decision variables

$T = \#$ of tables to produce

$C = \#$ of chairs to produce

Step 2 — objective function

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}, \quad (T, C) \mapsto 7T + 5C$$

Step 3 — constraints

$$\underset{T, C}{\text{maximize}} \quad 7T + 5C \quad \text{profit}$$

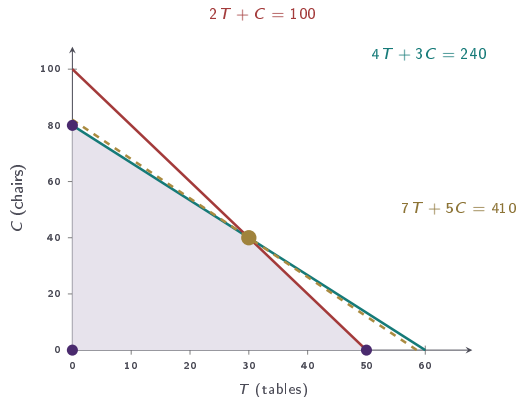
$$\text{subject to} \quad 4T + 3C \leq 240 \quad \text{carpentry hours}$$

$$2T + C \leq 100 \quad \text{painting hours}$$

$$T, C \geq 0$$

Two variables, so we can *draw* it. That is the point of starting here: everything true of this picture stays true in $\mathbb{R}^{500}$, where we cannot draw.

The feasible region is a polytope



Each constraint is a **halfspace**; their intersection is a **polyhedron**; bounded, it is a **polytope**. All three facts come from Chapter 1.

Corner	(T, C)	$7T + 5C$
A	$(0, 0)$	0
B	$(50, 0)$	350
C	$(30, 40)$	410
D	$(0, 80)$	400

At C: $4(30) + 3(40) = 240$ and $2(30) + 40 = 100$. *Both* constraints are binding — the optimum sits where two lines cross.

Why we only had to check four points

Corner point theorem

If an LP has an optimal solution, then at least one optimal solution occurs at an **extreme point** (vertex) of the feasible polytope.

Why. Chapter 1 established two facts:

1. A polytope is the *convex hull* of its extreme points: every feasible x is $x = \sum_i \lambda_i v_i$ with $\sum_i \lambda_i = 1$, $\lambda_i \geq 0$.
2. An affine function satisfies $f(\sum_i \lambda_i v_i) = \sum_i \lambda_i f(v_i)$ — *with equality*.

Hence $f(x)$ is a weighted average of the vertex values, and an average never exceeds the largest term:

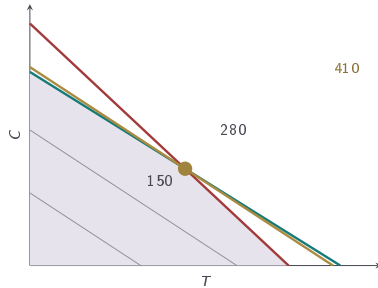
$$f(x) = \sum_i \lambda_i f(v_i) \leq \max_i f(v_i).$$

So no interior point can beat the best vertex. □

Takeaway

A continuum of candidates collapses to a *finite* list. This is the single idea behind the simplex method — and, in Part II, exactly the idea that integrality destroys.

The isoprofit line method



The objective $7T + 5C = k$ is a family of **parallel lines**, one for each profit level k , all with slope $-7/5$.

Push the line outward, parallel to itself, until it is about to leave the feasible region. The *last* point it touches is optimal.

Takeaway

This picture explains sensitivity analysis before we do any algebra: the optimum stays at $(30, 40)$ as long as the objective slope $-c_T/c_C$ stays between the two constraint slopes, $-4/3$ and -2 .

Four special cases

Infeasibility

No point satisfies all constraints; $\mathcal{F} = \emptyset$. *Example:* $x \geq 10$ and $x \leq 5$.

Fix: the constraints, not the solver.

Unboundedness

The objective improves without limit; $p^* = +\infty$. Almost always a missing constraint.

Takeaway

If the carpentry profit were \$6. $\bar{6}$ instead of \$7, the objective slope would be exactly $-4/3$ and every point on the edge from (30, 40) to (0, 80) would be optimal. Check it against the sensitivity report in three slides.

Redundancy

A constraint that does not touch the feasible region. Harmless for the answer, wasteful for the solver, and misleading when read as a binding business rule.

Alternative optima

The objective line is *parallel* to a binding constraint: an entire edge is optimal. Chapter 1 predicted this — the solution set of a convex problem is convex, so never two isolated points.

Sensitivity analysis — three questions

After solving, the interesting questions are not “what is the answer” but “how far can the world move before the answer changes”.

Shadow price

The change in the optimal objective value per *one-unit* increase in a constraint's right-hand side.

Zero for any non-binding constraint — you already have slack.

Reduced cost

For a variable at zero: by how much its objective coefficient must improve before it becomes worth using.

Allowable range

How far a coefficient or a right-hand side can move with the current basis — the current *corner* — staying optimal.

Where it breaks

A shadow price is valid *only* inside its allowable range. Extrapolating beyond it is the single most common error in LP-based decision memos: the binding constraint changes, and the price with it.

Sensitivity by hand, on the carpentry problem

Shadow price of painting hours. Raise the RHS from 100 to $100 + \Delta$:

$$\begin{cases} 4T + 3C = 240 \\ 2T + C = 100 + \Delta \end{cases}$$

Solving: $C = 240 - 2(100 + \Delta) \dots$ gives

$$T = 30 + 1.5\Delta, \quad C = 40 - 2\Delta.$$

Hence

$$Z = 7T + 5C = 410 + \underbrace{0.5}_{\text{shadow price}} \Delta.$$

One extra painting hour is worth 50 cents.

Range on the table profit c_T .

$(30, 40)$ stays optimal while the objective slope lies between the two constraint slopes:

$$\frac{4}{3} \leq \frac{c_T}{c_C} \leq 2 \implies 6.\bar{6} \leq c_T \leq 10.$$

With $c_T = 7$: allowable increase 3, allowable decrease $\frac{1}{3}$.

Takeaway

Every number Solver will print in two slides can be derived by hand from this picture. Do that once and the reports stop being magic.

Duality

Primal

$$\begin{array}{ll}\text{maximize} & {}^t c x \\ \text{subject to} & A x \leq b, \quad x \geq 0\end{array}$$

$$\begin{array}{ll}\text{max} & 7T + 5C \\ \text{s. t.} & \begin{pmatrix} 4 & 3 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} T \\ C \end{pmatrix} \leq \begin{pmatrix} 240 \\ 100 \end{pmatrix}\end{array}$$

How much to produce?

Dual

$$\begin{array}{ll}\text{minimize} & {}^t b y \\ \text{subject to} & {}^t A y \geq c, \quad y \geq 0\end{array}$$

$$\begin{array}{ll}\text{min} & 240y_1 + 100y_2 \\ \text{s. t.} & 4y_1 + 2y_2 \geq 7 \\ & 3y_1 + y_2 \geq 5\end{array}$$

What is an hour worth?

Takeaway

Note the transposition: the *rows* of the primal constraint matrix become the *columns* of the dual. One dual variable per primal constraint, one dual constraint per primal variable.

The two duality theorems

Weak duality

For any feasible x and any feasible y ,

$${}^t c x \leq {}^t b y.$$

Proof. ${}^t c x \leq {}^t ({}^t A y) x = {}^t y A x \leq {}^t y b$. Every dual feasible point is an *upper bound* on the primal. $\square$

Strong duality

If either problem has a finite optimum, both do, and the values are equal.

Check it

Solve the dual: $y_1 = 1.5$, $y_2 = 0.5$.

$$4(1.5) + 2(0.5) = 7 \quad \checkmark$$

$$3(1.5) + (0.5) = 5 \quad \checkmark$$

$$240(1.5) + 100(0.5) = 410$$

exactly the primal optimum.

Takeaway

The dual variables are the shadow prices: $y_1 = 1.5$ for carpentry, $y_2 = 0.5$ for painting. Duality and sensitivity are one subject.

Excel Solver

Excel Solver — laying out the sheet

The discipline: **one block for data, one row for the decision, one cell per constraint**. Never bury a coefficient inside a formula.

	C	D	E	F	G
3	profit / unit	7	5		
4		T	C	LHS	RHS
5	decision	30	40		=SUMPRODUCT(D3:E3,D5:E5)
6	carpentry	4	3	=SUMPRODUCT(D6:E6,\$D\$5:\$E\$5)	240
7	painting	2	1	=SUMPRODUCT(D7:E7,\$D\$5:\$E\$5)	100

G5 holds the *objective*; G6:G7 hold the *right-hand sides*.

Solver dialog

Set Objective: \$G\$5 To: Max

By Changing Variable Cells: \$D\$5:\$E\$5

Subject to: \$F\$6<=\$G\$6, \$F\$7<=\$G\$7

Make Unconstrained Variables Non-Negative ✓

Solving Method: **Simplex LP**

Where it breaks

Choose **Simplex LP**, not GRG Nonlinear, whenever the model is linear. GRG will usually find the same point but reports no shadow prices and gives no global guarantee. In Chapter 3 we switch to GRG deliberately — and pay attention to why.

Solver output 1 — the Answer Report

Objective Cell (Max)

Cell	Name	Original Value	Final Value
\$G\$5	profit	0	410

Variable Cells

Cell	Name	Original Value	Final Value	Integer
\$D\$5	T	0	30	Contin
\$E\$5	C	0	40	Contin

Constraints

Cell	Name	Cell Value	Formula	Status	Slack
\$F\$6	carpentry	240	\$F\$6<=\$G\$6	Binding	0
\$F\$7	painting	100	\$F\$7<=\$G\$7	Binding	0

Read it this way. Both constraints binding, zero slack — every hour of both resources is consumed. That is why the optimum sits at the intersection of the two lines, and why *both* shadow prices will be strictly positive.

Solver output 2 — the Sensitivity Report

Variable Cells

Cell	Name	Final Value	Reduced Cost	Obj. Coefficient	Allow. Increase	Allow. Decrease
\$D\$5	T	30	0	7	3	0.333
\$E\$5	C	40	0	5	0.25	1.5

Constraints

Cell	Name	Final Value	Shadow Price	R.H. Side	Allow. Increase	Allow. Decrease
\$F\$6	carpentry	240	1.5	240	60	40
\$F\$7	painting	100	0.5	100	20	20

Takeaway

These are the dual variables from two slides ago: $y_1 = 1.5$, $y_2 = 0.5$. And $c_T \in [7 - 0.333, 7 + 3] = [6.\bar{6}, 10]$ — exactly the range we derived from the slope condition.

Where it breaks

Carpentry hours are worth 1.5 each *only up to 300 hours* ($240 + 60$). Buy the 301st and the picture changes: painting becomes the sole bottleneck and the price collapses.

Workshop task 1 — reproduce it

20 minutes, at the keyboard

1. Build the carpentry sheet exactly as laid out. Use `SUMPRODUCT` throughout; no hard-coded numbers in formulas.
2. Solve with **Simplex LP**. Request all three reports (Answer, Sensitivity, Limits).
3. Confirm $T = 30$, $C = 40$, $Z = 410$, and both constraints binding.
4. Change the painting hours from 100 to 110. Predict the new profit *before* you re-solve, using the shadow price. Then re-solve.
5. Now change it to 130. Your prediction fails. **Explain why** — the allowable increase was 20.
6. Set the table profit to $\$6.\overline{6}$ and re-solve. What happened, and which special case is this?

Item 5 is the point of the exercise. Item 6 is the one that appears on the exam.

Portfolio selection as a linear program

Fil rouge — NEOMAMA Brokerage

Fil rouge — NEOMAMA Brokerage

\$250,000 to invest. The client's guidelines: **(a)** municipal bonds at least 20% of the investment; **(b)** at least 40% in a combination of electronics, aerospace and drugs; **(c)** the nursing-home stock at most 50% of the amount in municipal bonds. Subject to these, maximize projected return.

	Investment	Return
X_1	LA municipal bonds	5.3%
X_2	Thompson Electronics	6.8%
X_3	United Aerospace	4.9%
X_4	Palmer Drugs	8.4%
X_5	Happy Days Nursing Homes	11.8%

X_j = dollars placed in investment j .

$$\max \quad 0.053X_1 + 0.068X_2 + 0.049X_3 + 0.084X_4 + 0.118X_5$$

$$\text{s. t. } X_1 + X_2 + X_3 + X_4 + X_5 = 250,000$$

$$X_1 \geq 0.20(250,000) \quad (\text{a})$$

$$X_2 + X_3 + X_4 \geq 0.40(250,000) \quad (\text{b})$$

$$X_5 \leq 0.50X_1 \quad (\text{c})$$

$$X_j \geq 0$$

Note (c). "At most half of what is in municipal bonds" is *not* a constant right-hand side. Rewrite it as $X_5 - 0.5X_1 \leq 0$ before it goes anywhere near Solver.

The answer, and what it teaches

Optimal allocation

	Amount	Share
Municipal bonds	\$50,000	20%
Thompson Electronics	\$0	0%
United Aerospace	\$0	0%
Palmer Drugs	\$175,000	70%
Nursing Homes	\$25,000	10%
Return	\$20,300	8.12%

Read the solution, don't just report it

- Constraint (a) is binding at exactly 20%. Municipal bonds return only 5.3%, so the model buys the *minimum* the client allows.
- But (a) is not pure cost: it *unlocks* the nursing-home stock through (c). Each municipal dollar buys 50 cents of 11.8% exposure.
- Net effect: $0.053 + 0.5(0.118) = 0.112$ per municipal dollar against 0.084 for Palmer. Yet the optimum still sits at the floor — because *both* dollars come out of Palmer:
 $0.112 - 1.5(0.084) = -0.014 < 0$.
- Electronics and Aerospace are dominated by Palmer Drugs and never enter.

Where it breaks

An LP will always concentrate: the optimum sits on a vertex, so it holds at most as many assets as there are binding constraints. **Nothing here penalises concentration risk** — there is no covariance in the model. That absence is precisely what Chapter 3 repairs.

Two more portfolio LPs

Texas Lotto — \$50,000

Petrochemical: 12% return, risk index 8.

Public utility: 6% return, risk index 4.

Average risk index must not exceed 6.

$$\max 0.12P + 0.06U$$

$$\text{s. t. } P + U = 50,000$$

$$\frac{8P+4U}{50,000} \leq 6$$

$$P, U \geq 0$$

Optimum: $P = U = \$25,000$, return \$4,500 (9%), risk exactly 6.

The same client, other way round

Now minimize risk, subject to earning at least 8%.

$$\min 8P + 4U$$

$$\text{s. t. } P + U = 50,000$$

$$0.12P + 0.06U \geq 4,000$$

$$P, U \geq 0$$

Optimum: $P = \$16,667$, $U = \$33,333$, average risk 5.33.

Takeaway

Two objectives, two problems, two answers — and a whole *frontier* of answers in between, one for each risk tolerance. Holding one objective as a constraint while optimizing the other is called the **ϵ -constraint method**. It is the backbone of the multi-objective chapter.

Audit the machine — a formulation that looks better

The same NEOMAMA brief was given to an AI assistant, which produced a complete model and solved it.

AI assistant — response

$$\max 0.053X_1 + 0.068X_2 + 0.049X_3 + 0.084X_4 + 0.118X_5$$

$$\text{s. t. } X_1 + X_2 + X_3 + X_4 + X_5 = 250,000$$

$$X_1 \geq 50,000 \quad (\text{a})$$

$$X_2 + X_3 + X_4 \geq 100,000 \quad (\text{b})$$

$$X_5 \leq 125,000 \quad (\text{c) note the RHS}$$

$$X_j \geq 0$$

Solver result. $X_1 = 50,000$, $X_4 = 100,000$, $X_5 = 100,000$.

Projected return \$22,850, or 9.14% — comfortably ahead of the client's requirements. All constraints satisfied.

Your audit — 25 minutes

1. Re-read guideline (c) in the client brief. Is constraint (c) a faithful translation?
2. Write the correct constraint. Why is its right-hand side *not* a constant?
3. Re-solve in Solver. What is the true optimal return?
4. How much apparent return did the error manufacture?
5. What happens to the *nursing-home position* as a share of the portfolio, and why does that matter more than the return?

Where it breaks

Everything here is internally consistent: the algebra is right, Solver ran correctly, the reported solution really does satisfy the stated constraints. **The model is simply not the client's problem** — and it errs in the flattering direction, which is exactly the kind of error nobody questions.

Part II — Combinatorial optimization

What changes when fractions are forbidden

Drop assumption 4 — divisibility — and require $x_j \in \{0, 1\}$. Three things break at once.

Convexity

$\{0, 1\}^n$ is a set of isolated points. The midpoint of two feasible solutions is infeasible — exactly the counterexample from Chapter 1.

Local = global

The theorem needed a convex feasible set. Without it, a locally best solution proves nothing about the global optimum.

Vertices

The corner point theorem is gone. There is no continuum to collapse — only a finite but *astronomically large* list.

Takeaway

“Finite” sounds easier than “infinite”. It is not. The LP over a polytope in $\mathbb{R}^{500}$ solves in milliseconds; the same problem with $x \in \{0, 1\}^{500}$ can outlive the universe. Part II is about why, and what to do about it.

The project selection problem

A fund can back projects $P_1, \dots, P_7$. Project P_i costs c_i and returns utility u_i . The total budget is $B = 45$.

$$x_i = \begin{cases} 1 & \text{if } P_i \text{ is selected,} \\ 0 & \text{if not.} \end{cases}$$

$$\max \sum_{i=1}^7 u_i x_i$$

$$\text{s. t. } \sum_{i=1}^7 c_i x_i \leq B$$

$$x_i \in \{0, 1\}$$

This is the **0–1 knapsack problem**.

	Utility u_i	Cost c_i	Ratio u_i/c_i
P_1	7	3	2.33
P_2	4	4	1.00
P_3	10	15	0.67
P_4	8	6	1.33
P_5	12	20	0.60
P_6	9	12	0.75
P_7	3	1	3.00
Total	53	61	

Total cost 61 against a budget of 45: we must leave out at least 16 units' worth. *Which?*

Takeaway

A solution is a binary string, e.g. $(1, 1, 1, 1, 0, 0, 1)$. There are $2^7 = 128$ of them.

Complete enumeration — and the wall

With n projects there are 2^n subsets. For $n = 7$ we could list all 128 by hand. Now scale up.

n	2^n subsets	Time at 10^9 subsets per second
10	1.0×10^3	instant
20	1.0×10^6	1 millisecond
30	1.1×10^9	1 second
40	1.1×10^{12}	18 minutes
50	1.1×10^{15}	13 days
60	1.2×10^{18}	37 years
100	1.3×10^{30}	4×10^{13} years — 2,900 times the age of the universe

Where it breaks

A European equity universe has some 600 names. Choosing a subset of them is a 2^{600} problem. No improvement in hardware touches this: doubling computer speed buys you *one* extra asset.

P, NP, and NP-hard — the vocabulary

Complexity theory classifies *decision* problems, whose answer is yes or no.

P — solvable

Problems solvable in time polynomial in the input size, $O(n^k)$.

Examples: linear programming, shortest path, sorting, matrix inversion.

NP — checkable

Problems whose yes answer can be *verified* in polynomial time, given a certificate.

Knapsack: hand me a subset; I add up costs and utilities in $O(n)$ and confirm. Verifying is easy. *Finding* is the hard part.

NP-hard — at least as hard

A problem H is NP-hard if *every* problem in NP reduces to it in polynomial time. Solve H efficiently and you solve all of NP.

NP-complete

In NP *and* NP-hard. The hardest problems that are still checkable.

$P \subseteq NP$. Whether $P = NP$ is open — one of the seven Clay Millennium problems, unsolved since 1971.

Knapsack is NP-hard

Decision version

Knapsack: given costs c_i , utilities u_i , budget B and a target K , is there a subset S with $\sum_{i \in S} c_i \leq B$ and $\sum_{i \in S} u_i \geq K$?

This was among Karp's original 21 NP-complete problems (1972). The reduction runs $\text{Subset-Sum} \leq_p \text{Knapsack}$: set $u_i = c_i$ and $K = B$, and the knapsack question becomes “is there a subset summing exactly to B ?” — and Subset-Sum is itself NP-complete.

The *optimization* version — our project selection — is therefore **NP-hard**: an efficient optimizer would answer the decision version too.

What NP-hard does and does not mean

- **It does not mean** your instance is unsolvable. Solvers routinely crack knapsacks with thousands of items.
- **It does not mean** no algorithm exists. Enumeration always works.
- **It does mean** no algorithm is known that is guaranteed polynomial on every instance — and that finding one would settle $P = NP$.
- **Practically**: expect exponential worst cases, and design around them with bounds, relaxations and heuristics.

An honest footnote: the $O(nB)$ algorithm

Knapsack *does* have a dynamic programming algorithm. Let $V[i, w]$ be the best utility using only items $1, \dots, i$ within budget w :

$$V[i, w] = \max \left\{ \underbrace{V[i-1, w]}_{\text{leave } P_i}, \underbrace{V[i-1, w - c_i] + u_i}_{\text{take } P_i} \right\}, \quad V[0, w] = 0.$$

Filling the table costs $O(nB)$ — for $n = 7$, $B = 45$: 315 cells. Trivial.

So why is it still NP-hard?

Because $O(nB)$ is polynomial in the *value* B , not in the *size of the input*. Writing B down takes $\log_2 B$ bits, so $B = 2^{\log_2 B}$ is exponential in the input length. An instance with $n = 100$ items and $B = 10^{18}$ has a tiny description and an impossible table.

This is called a **pseudo-polynomial** algorithm. It is the reason knapsack feels easy in the classroom and hard in production.

The greedy heuristic

Sort by **utility per unit of cost** u_i/c_i and take items while the budget allows.

Order		u_i/c_i	Cum. cost	Cum. utility
1	P_7	3.00	1	3
2	P_1	2.33	4	10
3	P_4	1.33	10	18
4	P_2	1.00	14	22
5	P_6	0.75	26	31
6	P_3	0.67	41	41
7	P_5	0.60	61	— stop

Greedy solution (1,1,1,1,0,1,1): cost 41, utility **41**, with 4 units of budget left unused and P_5 too expensive to fit.

Takeaway

On *this* instance greedy happens to be optimal. Drop P_5 (utility 12) and you keep 41; any other way of freeing 16 units of budget costs at least 13 in utility. So $P^* = 41$.

But greedy has no guarantee

Budget $B = 10$, three items:

	u	c	u/c
A	7	6	1.17
B	5	5	1.00
C	5	5	1.00

Greedy takes A (cost 6), then neither B nor C fits: value **7**. Optimum is $B + C$: value **10**. Greedy is 30% off.

Relaxation — bounding what you cannot compute

The integer problem

$$\begin{aligned} P^* = \max \quad & \sum_i u_i x_i \\ \text{s. t.} \quad & \sum_i c_i x_i \leq B \\ & x_i \in \{0, 1\} \end{aligned}$$

Its LP relaxation

$$\begin{aligned} P_R = \max \quad & \sum_i u_i x_i \\ \text{s. t.} \quad & \sum_i c_i x_i \leq B \\ & 0 \leq x_i \leq 1 \end{aligned}$$

The relaxation is an *LP* — Part I solves it.

The sandwich

$$P_0 \leq P^* \leq P_R$$

Left: any feasible integer solution x^0 gives a **lower bound**
 $P_0 = \sum_i u_i x_i^0$.

Right: the relaxation has a larger feasible set, so its optimum is an **upper bound**.

On our instance

Fill greedily allowing fractions: after P_3 we have cost 41, utility 41, and 4 units left. Take $4/20$ of P_5 : $+0.2(12) = 2.4$.

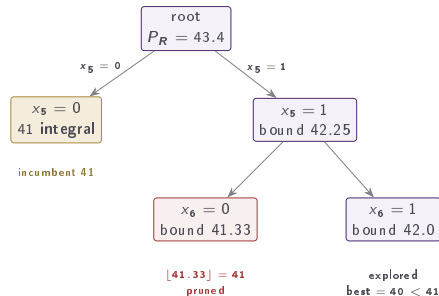
$$P_R = 43.4 \implies 41 \leq P^* \leq \lfloor 43.4 \rfloor = 43.$$

A certificate that greedy is within 5% — obtained without solving anything integer.

Branch and bound

The bounds turn into an exact algorithm.

1. **Relax.** Solve the LP. If the solution is integral, stop.
2. **Branch.** Pick a fractional x_j and split into $x_j = 0$ and $x_j = 1$.
3. **Bound.** Solve each relaxation for an upper bound.
4. **Prune.** If a node's bound cannot beat the best integer solution found so far — the *incumbent* — discard the whole subtree unexplored.
5. Repeat until every node is solved or pruned.



Five nodes settle it: $P^* = 41$, against $2^7 = 128$ subsets.

Takeaway

All u_i are integers, so any bound may be rounded *down*: a node with bound 41.33 can yield at most 41 — no better than the incumbent. That rounding is what prunes the left branch below.

Reading the tree

Node	Fixed	LP bound	What happens
root	—	43.4	$x_5 = 0.2$ is fractional; branch on x_5 .
1	$x_5 = 0$	41	The remaining six projects cost $41 \leq 45$, so the LP takes all of them. Integral — incumbent $Z = 41$.
2	$x_5 = 1$	42.25	$\lfloor 42.25 \rfloor = 42 > 41$: cannot prune. $x_6 = 11/12$ is fractional; branch on x_6 .
3	$x_5 = 1, x_6 = 0$	41.33	$\lfloor 41.33 \rfloor = 41$, no better than the incumbent. Pruned without exploring.
4	$x_5 = 1, x_6 = 1$	42.0	$42 > 41$: explored. Best integer solution here is $\{P_1, P_2, P_4, P_5, P_6\}$ with utility 40. No improvement.

Takeaway

Node 2 is the honest part of the story. Its bound of 42.25 did *not* prune, and exploring it cost two more nodes to learn nothing. Bounds prune *often*, not always — which is exactly why the worst case stays exponential.

Modelling with 0–1 variables

Fil rouge — Trapeze Investments

A venture capital firm is evaluating six investment opportunities. There is not enough capital for all of them, but more than one will be chosen. Let $x_i = 1$ if opportunity i is selected. Write the constraint for each rule.

Business rule	Constraint
(a) At least 3 of the six are selected	$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 \geq 3$
(b) Either 1 or 4, but not both	$x_1 + x_4 = 1$
(c) If 4 is selected then 6 must be; 6 may stand alone	$x_4 \leq x_6$
(d) 5 cannot be selected unless both 2 and 3 are	$2x_5 \leq x_2 + x_3$
(e) 5 must be selected if both 2 and 3 are	$x_5 + 1 \geq x_2 + x_3$

Takeaway

Note (d) and (e) are *converses*, and their constraints are not each other's mirror image. Check every logical constraint by tabulating all four combinations of (x_2, x_3) — it takes thirty seconds and catches most errors.

The portfolio rules that need 0–1 variables

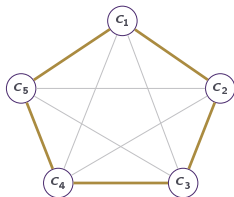
Back to NEOMAMA. Let X_j be dollars in asset j and $y_j \in \{0, 1\}$ a “do we hold it at all” indicator, with M a large constant.

Rule the client actually states	Constraint	
“Hold at most K different assets”	$\sum_j y_j \leq K$	<i>cardinality</i>
“If you hold it at all, hold at least \$10,000”	$10,000 y_j \leq X_j \leq M y_j$	<i>semi-continuous</i>
“Bonds and the nursing home are mutually exclusive”	$y_1 + y_5 \leq 1$	
“Buy the aerospace ticket whole or not at all”	$X_3 = 75,000 y_3$	

Where it breaks

Each of these is a sentence any portfolio manager would consider unremarkable. Each of them turns a problem that solves in milliseconds into an NP-hard one. **This is the central practical lesson of the chapter:** the cost of a constraint is not measured in basis points of return but in complexity class.

Beyond knapsack — the travelling salesman



A tour visits every city once and returns.

With 5 cities there are $5! = 120$ orderings ($4!/2 = 12$ genuinely distinct tours).

Tour	Length
$C_1 - C_2 - C_3 - C_4 - C_5 - C_1$	29
$C_1 - C_2 - C_4 - C_5 - C_3 - C_1$	59

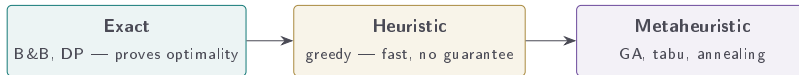
In general $(n-1)!/2$ distinct tours. For $n = 20$ that is 6×10^{16} — and $n!$ grows faster than 2^n .

Takeaway

TSP is NP-hard too. In finance the same structure appears as *optimal execution order* and *index rebalancing with round-lot constraints*.

When exact methods run out: a first look at metaheuristics

Three tiers of method, in increasing order of what you give up:



Genetic algorithm, in six steps

1. Encode the problem — create the chromosomes
2. Select an initial population
3. Evaluate fitness; duplicate the fit
4. Crossover and mutation
5. Form the new population
6. Stop on a criterion

Chromosomes on our instance

$(1, 0, 0, 0, 1, 0, 1) \rightarrow 7 + 12 + 3 = 22$
 $(0, 1, 1, 0, 0, 1, 1) \rightarrow 4 + 10 + 9 + 3 = 26$
 $(1, 1, 1, 1, 1, 1, 1) \rightarrow -\infty$ infeasible: cost $61 > 45$

Fitness = utility, with infeasible chromosomes penalised. Chapter 6 develops this properly.

Problems

Problems — set A: linear programming

1. **Bhavika Investments.** \$200,000 must go into a stock fund S and a money market fund M ; the annual return must be at least \$14,000; at least \$40,000 must sit in the money market fund. Risk is $12S + 5M$.

$$\min 12S + 5M \quad \text{s.t.} \quad S + M = 200,000, \quad 0.10S + 0.05M \geq 14,000, \quad M \geq 40,000.$$

- (a) Solve graphically. (b) What is the total return, and the rate of return? (c) For each additional dollar available, how much does risk change? (d) Would the solution change if the money-market floor were \$50,000?
2. **Texas Lotto, revisited.** Solve both formulations from the earlier slide graphically. Plot the two optima on the same (risk, return) plane. Then find three more points by varying the return floor. What have you just drawn?
 3. **NEOMAMA sensitivity.** Solve the NEOMAMA LP in Solver. The municipal-bond floor is binding. (a) What is its shadow price? (b) Interpret its sign in one sentence for the client. (c) Over what range of the floor does that price hold?
 4. **Duality by hand.** Write the dual of the NEOMAMA LP. How many dual variables? How many dual constraints? Verify strong duality numerically against your Solver output.

Problems — set B: combinatorial optimization

5. **Project selection.** For the seven-project instance with $B = 45$: (a) confirm the greedy solution by hand; (b) compute the LP relaxation bound P_R ; (c) prove that $P^* = 41$ by showing no exclusion set of cost ≥ 16 has utility below 12.
6. **Budget sensitivity.** Re-solve the same instance with $B = 40$ and $B = 50$. Plot optimal utility against B . Is the curve concave? Is it *smooth*? Contrast with the LP shadow price of Part I.
7. **Greedy's blind spot.** Construct a knapsack instance on which greedy achieves less than half the optimum. Prove your bound.
8. **Logical constraints.** Write 0–1 constraints for: (a) at most two of $\{1, 2, 3\}$; (b) if 1 and 2 are both chosen then 3 must be; (c) exactly one of $\{4, 5, 6\}$; (d) 3 is chosen only if at least two of $\{1, 2, 4, 5\}$ are.
9. **Cardinality on the fil rouge.** Add to the NEOMAMA LP: at most three assets may be held, and any held position must be at least \$20,000. Formulate with 0–1 variables and solve in Solver (Simplex LP handles binaries). Compare with the pure LP answer. By how much did the return fall? What did the client buy with it?
10. **Synthesis.** Explain, in five sentences and without formulas, to a portfolio manager who has never heard of complexity theory, why “just hold your best 25 names” is a fundamentally harder request than “maximize return subject to sector limits”.

Summary

Part I — linear programming

- An affine objective over a polytope: the optimum sits at a vertex.
- Four special cases: infeasible, unbounded, redundant, alternative optima.
- Sensitivity: shadow prices, reduced costs, allowable ranges — valid only inside their range.
- Duality: one dual variable per constraint; the dual optimum *is* the vector of shadow prices.
- Solver: lay the sheet out properly, choose Simplex LP, read all three reports.

Part II — combinatorial optimization

- Integrality destroys convexity, and with it the vertex argument.
- 2^n subsets: a wall no hardware crosses.
- Knapsack is NP-hard; its $O(nB)$ DP is pseudo-polynomial, not a counterexample.
- Greedy is fast and unguaranteed; relaxation gives $P_0 \leq P^* \leq P_R$.
- Branch and bound turns those bounds into an exact method.
- 0–1 variables express the rules clients actually state — at a price.

Takeaway

NEOMAMA's portfolio is now an LP we can solve and an IP we can bound. What it still lacks is any notion that two assets can move together. **Chapter 3** replaces the linear risk index with $\mathbf{w}^T \Sigma \mathbf{w}$, and the corner point method with gradients and Newton steps.

Reading. Full references, with what each source is for, on the next slide.

References

References and further reading

Core

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- Karp, R. M. (1972). Reducibility among combinatorial problems. In: *Complexity of Computer Computations*, Plenum, 85–103. the original 21 problems, including Subset-Sum and Knapsack.
- Garey, M. R. & Johnson, D. S. (1979). *Computers and Intractability*. Freeman.
- Wolsey, L. A. (2020). *Integer Programming*, 2nd ed. Wiley. why a tighter formulation beats a faster solver.

Metaheuristics for the multi-objective knapsack

- [BA] Zouache, D., Moussaoui, A. & Ben Abdelaziz, F. (2018). A cooperative swarm intelligence algorithm for multi-objective discrete optimization with application to the knapsack problem. *European J. Operational Research* 264(1), 74–88.

Takeaway

Next. Chapter 3 — nonlinear optimization: Taylor, gradient and Hessian, descent methods, Newton, and the mean–variance portfolio.